

PRIVATE RENTING ANALYSIS OF ENGLAND AND WALES (2011 -2021)

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Abstract

The present study is a visual analytics exploration into how educational qualifications, employment, and housing tenure are interlinked, covering 323 Local Authority Districts (LADs) in England and Wales and based on the 2011 and 2021 census data. The main theory suggests that the level of qualification can be a reliable indicator of both unemployment and private renting, and that areas with larger shares of Level 4 and above qualifications will display the biggest changes to private renting over the decade. Four census tables were combined, cleaned and analysed through data visualisation in Tableau and Principal Component Analysis (PCA) and t-distributed Stochastic Neighbour Embedding (t-SNE) for dimensionality reduction. Bayesian linear regression was further employed to forecast private renting rates to 2030 with credible intervals. The main results show a mild positive association ($r = 0.41$) between Level 4 qualifications and private renting in 2021, and a very strong inverse association with unemployment ($r = -0.39$). London boroughs are identified as standing out throughout: the City of London is recording 68.4% Level 4 attainment and private renting percentage which have increased from 35.9% in 2011 to 48.3% in 2021. Bayesian analysis estimates that the national average private renting will increase to about 22.8% by 2030, while LADs with a high level of qualification are anticipated to be the ones with the most significant increases. These discoveries very much affect housing and labour market policies, planning, and studies on spatial inequality.

1. Introduction

How educational level, economic activity, and housing forms affect each other and relates to socio-economic inequality in present day Britain is a highly debated topic. After the global financial crisis in 2008, the private rented sector (PRS) in England and Wales was rapidly enlarged so that its share of the housing system more than doubled and many working-age adults have now choosing renting instead of owner-occupation as their housing tenure [1]. This rise of 'Generation Rent' phenomenon has been identified and described in literature. However, it is not spread across the whole country but rather very strongly concentrated in the areas with high concentrations of graduate workers and especially in London and other big urban labour markets where the prices and costs are highest [2].

Moreover, educational attainment is nowadays a sort of a dividing line in the labour market where people with Level 4 qualifications and above (degree-level and equivalent) are almost always safer from unemployment, earning more money and are also more likely to move to a different place of residence [3]. The 2011 Census gives a very detailed cross-sectional view of these relationships at the Local Authority District level before the period of austerity, welfare reform and structural changes in the housing market when these started to take place. The 2021 Census which was taken during the COVID-19 pandemic times, is a chance to look at these structural pattern changes in depth i.e. to see if these have intensified, changed location or even become different in unexpected ways.[4]

This project is designed to be useful to a range of end users, including urban planners and housing policy analysts at local authorities who need to understand which areas face housing pressures driven by graduate population growth; regional economists tracking labour market participation and skill utilisation; and researchers studying spatial inequality and the socio-economic geography of England and Wales. The interactive Tableau dashboard is specifically designed to allow non-specialist users to explore the data by geography, filter by qualification or tenure variable, and draw their own conclusions, while the Python-based dimensionality reduction and forecasting outputs provide deeper analytical structure.

The hypothesis underpinning this work is: qualification level is a strong predictor of both employment status and housing tenure in 2011, and regions with higher Level 4+ qualifications show greater shifts toward private renting by 2021. This report documents the full analytical lifecycle from data selection and preparation through task definition, visualisation design, and a Bayesian trend projection to 2030.

London boroughs and many parts of the South of England remain highly visible throughout the different stages of the analysis, it should be noted. London not only leads in qualification and private renting by a considerable margin, but it also keeps popping up in the principal component analysis (PCA) and t-SNE projections as well as in the Bayesian forecasts. The South East, which includes university towns and well-to-do suburbs with a high number of graduates, forms a cluster with London in multivariate space. This is probably one of the reasons why these areas visually stood out in the last dashboard, isn't it!

2. Data Preparation and Abstraction

2.1 Dataset Selection

We chose four census datasets to define the main socio-economic aspects of the hypothesis. To be specific, from the 2011 Census, the data on Economic Activity (table QS601EW) gave us information about people who are working and looking for work aged 16-74 per LAD level. Tenure (table QS405EW) showed us the number of households in each tenure category including private renting. Highest Level of Qualification (table QS501EW) gave us data about the percentage of the resident population at each qualification level ranging from no qualifications to Level 4 and above. And for the 2021 Census, the same Tenure dataset (TS054) was downloaded from the ONS Nomis bulk portal which allowed to compare private renting rates directly over the decade.

Since the boundaries of Local Authority Districts changed between 2011 and 2021 (some districts were even merged like in Somerset and Northamptonshire), geographic harmonisation had to be done. ONS 'Change Over Time' geographic lookup tool helped to match LADs between the two census years, and LADs that were restructured were either combined or left out to make sure the data was comparable. In the end, the final merged dataset consisted of 323 LADs for England and Wales.

2.2 Data Cleaning and Derived Variables

The four datasets were first imported into Python using pandas. The merging operation was based on the shared area code. To enable meaningful proportion comparisons between LADs of vastly different population sizes, several new variables were created. Among them, the unemployment rate was defined as the number of unemployed persons divided by the total economically active population aged 16-74. The percentages of the private rented households in 2011 and 2021 were obtained by dividing the private rented household figures by the total household figures respectively. Likewise, the percentages of residents with Level 4 and above qualifications and those without any qualifications were calculated using the respective raw counts.

A comprehensive missing value check was performed on the combined dataset. Most importantly, no missing values were found in any of the 16 variables in the final merged dataset ($n = 323$). The excellent coverage rate of ONS census data accounts for this situation. However, it should be mentioned that the geographic harmonisation process involved careful handling of boundary changes. Thus, several LADs that had been extensively restructured were handled by population-weighted aggregation rather than omission.

2.3 Data Types and Semantics

The dataset is made up of both quantitative ratio variables (proportions, rates) and categorical identifiers (area name, area code). The main variables of analysis are all continuous, ratio-scale quantities which are also bounded between 0 and 1. This makes them very versatile as they can be directly compared with each other, used in correlation analysis or as predictors in linear models.

The territorial level chosen for the unit of analysis is the Local Authority District. This level offers a compromise between regional level and output-area analysis: Major variables used in the analysis include:

pct_level4 (percentage of regular residents aged 16 and over with Level 4+ qualifications), unemployment_rate (percentage of economically active population that is unemployed), pct_private_rented_2011 and pct_private_rented_2021 (household tenure proportions), and pct_no_qual (percentage of persons with no formal qualifications). These five factors delineate the main analytical framework for all the further visualization and modelling stages.

3. Task Definition

Following Munzner's task taxonomy [5], the analytic tasks addressed in this project can be characterised across three levels: high-level actions, mid-level search tasks, and low-level query targets.

3.1 High-Level Tasks

Mainly, the first high-level task is Discover: the visualisation aims at uncovering the hidden or unverified relationships, between education level, unemployment, and housing situation. The second task is Present: The Tableau dashboard has been created with the purpose of advising policy analysts about the major insights in a friendly and interactive mode. Users can also validate or refute the main hypothesis by hands-on exploring the data.

3.2 Mid-Level Search and Query Tasks

Several user requirements resulted in the identification of mid-level tasks. The first is Compare: users want to compare the private renting rates of LADs between 2011 and 2021 in order to find out which areas have undergone the greatest changes in tenure. Second, Lookup: users might want to locate specific LADs, for example their own local authority, and get values of individual socio-economic indicators. Third, Browse: the scatter plots and outputs of dimensionality reduction are meant to be used as a support for browsing activity so that users can find clusters and outliers without having a specific LAD in mind. Fourth, Summarise: the choropleth map display is helpful in identifying regional patterns in the overall distribution of each variable.

At the lowest level, the precise aims are: first, finding out how pct_level4 and pct_private_rented values vary among different LADs; second, examining the changes and the relationship between qualification level and tenure change; third, spotting the anomalies in PCA and t-SNE plots; and finally, the ability to click on a specific LAD in the visualization and bring up its complete socio-economic profile. Besides serving as examples, these tasks guided the selection of the visualization techniques illustrated in the next section.

4. Visualisation Justification

4.1 Choropleth Map

We employed a choropleth map as the main geographical presentation of the distribution of pct_level4 and pct_private_rented_2021 across England and Wales. In fact, a choropleth map is a suitable choice here since the data is numerical and the spatial unit used (Local Authority District) is a geographic polygon with no inherent order. Light-to-dark hues in the sequential colour scheme are a great way to represent the magnitude of a single variable, as they make use of the human perceptual ability to distinguish different levels of brightness [5]. To ensure proper polygon placement, custom geocoding was performed in Tableau using the LAD boundary data from ONS.

The use of a diverging colour scheme was considered for the difference variable (change in private renting 2011 to 2021), as diverging schemes are appropriate when a meaningful midpoint (zero change) exists. However, because 322 of the 323 LADs showed an increase in private renting over the decade, a sequential scheme from pale yellow to dark orange was ultimately used for change variables, as there was minimal variance below zero.

4.2 Scatter Plots and Trend Lines

Scatter plots were a principal medium of analysis for visualizing pairwise relations of core variables. The scatter plot idiom encodes two quantitative attributes by means of the two most perceptually accurate channels: horizontal and vertical position [5]. To make it easier to detect directional relations, linear trend lines with 95% confidence intervals (CI) bands were added. A scatter plot of `pct_level4` against `pct_private_rented_2021` is at the heart of the inquiry, with each point representing an LAD and size conveying the total number of households as a way of population context. Tooltip capability in Tableau makes it possible for users to obtain the complete name of the LAD and variable values when the mouse pointer is hovered over the mark, thus meeting the Lookup task.

4.3 Bar Charts and Comparative Views

Ranked bar charts were used to support direct comparison of LADs on single variables such as percentage change in private renting. The bar chart idiom is optimal for comparing a single quantitative attribute across many categorical items, as it uses position on a common aligned scale, the most accurate encoding channel for quantitative comparison [5]. A dual-axis bar chart was used to show 2011 and 2021 private renting rates side by side for the top and bottom LADs by qualification level, enabling direct intercensal comparison.

4.4 PCA Projection

Principal Component Analysis was conducted on the five main quantitative variables (`pct_level4`, `pct_no_qual`, `unemployment_rate`, `pct_private_rented_2011`, `pct_private_rented_2021`), first normalized to a mean of zero and a standard deviation of one. PCA is a method of linear dimensionality reduction that finds new axes which are uncorrelated and capture the most variance in the data [5]. The top two principal components (PC1 and PC2) were extracted as a CSV and visualised in Tableau as a scatter plot, with colour mapped to `pct_level4` and tooltip showing the names of areas.

PCA was the first projection technique used here since it is understandable: the principal components are basically new features obtained by combining the old ones, and the loadings can be analysed to see which variables contribute most to each component. This level of openness is very necessary when reporting results to policymakers who may be not very much acquainted with non-linear embedding techniques [6]. The PCA projection displays a range from highly qualified, highly renting London boroughs at one extreme to poorly qualified, low renting northern English and Welsh LADs at the other end, with unemployment forming an opposite gradient.

4.5 t-SNE Projection

t-distributed Stochastic Neighbour Embedding or shortly t-SNE was implemented by van and Maaten [6] as a second method of non-linear projections. t-SNE works by minimizing Kullback-Leibler divergence between two pairwise similarity distributions - one in the high dimensional and the other in the low dimensional space resulting that similar points are close to each other in the final projection. Contrary to Principal Component Analysis, t-SNE is totally awesome in finding local clusters of data points that remain hidden in singular global linear variance decomposition.

For this work, t-SNE has been executed with perplexity = 30 and 1000 iterations on the identical five-variable standardised input space. The production (`tSNE_1`, `tSNE_2`) was saved and displayed in Tableau with colour indicating `pct_level4`, size by `unemployment_rate`, and tooltips presenting the full area name and all five variables. The t-SNE mapped out clusters: a tight group of London boroughs, a fuzzy group of English provincial LADs, and a loose group of Welsh LADs, with very high no-qualifications and low rates of private renting. Both types of mappings fulfill their purposes complementarily: PCA shows the overall structure of data and provides interpretable loadings while t-SNE shows very local cluster structures. Relying on the two techniques is actually the best way to do exploratory data analysis, according to [7].

4.6 Bayesian Trend Forecasting

Bayesian linear regression was employed to forecast private renting rates for each LAD to 2030, using `pct_level4`, `pct_no_qual`, and `unemployment_rate` as predictors and `pct_private_rented_2021` as the response variable. The Bayesian framework was chosen because it provides not merely point estimates but

full posterior distributions over model parameters, enabling the computation of credible intervals for 2030 forecasts [8]. This is particularly valuable in a policy context, where understanding uncertainty ranges is as important as the central prediction itself.

Regression coefficients were assigned weak, uninformative priors ($\text{Normal}(0, 10)$), which is in line with a principled Bayesian perspective that permits the data to have the main say in the inference. 95% credible intervals of a LAD's predicted private renting rate in 2030 were generated using the posterior predictive distribution. In Tableau, these were displayed as error bars on a ranked point plot; thereby users can view not only the LADs with the highest forecast uncertainty but also those with the highest predicted values.

5. Evaluation

5.1 Evaluation Framework

We validated the visualisation thoroughly using Munzner's four-level validation framework (domain abstraction idiom, algorithm) [5]. For the domain level, the task definitions were backed by a literature review of academic sources on housing tenure and educational attainment, which established that the choice of variables is consistent with researchers and policy analysts main instruments in this field [1][3]. At the level of abstraction, the choice of using proportions rather than absolute counts was justified by comparing the scatter plot with population-weighted and unweighted views: the proportional view corrected the systematic bias that the large population size of London boroughs had caused.

5.2 Analytic Evaluation

An analytic evaluation was planned as a means of gauging the effectiveness of the dashboard through peer discussion groups. These are the tasks put forward for the evaluation session: Initially, a simple lookup task requesting participants to find the private renting rate of a particular LAD (e.g. Darlington) in both 2011 and 2021; secondly, a comparison task requiring participants to find the five LADs with the highest predicted private renting rates in 2030; thirdly, a pattern-finding task requiring participants to explain what the PCA scatter plot indicates about the relationship between qualification level and tenure; and finally, a synthesis task asking participants whether they think the visualisation is supportive of or contradicts the main hypothesis.

Due to constraints on the availability of discussion group members, the evaluation was not conducted in a live session. Instead, a cognitive walkthrough of the dashboard was performed using these tasks as the evaluation lens. So, choropleth map was immediately understandable for pattern-finding task; London's very high pct_level4 together with high private renting rates were visually very clear even without tooltip interaction. More chart-guided interaction was necessary for the PCA scatter plot: without the tooltip labels, the cluster of London boroughs in the high-PC1 region wasn't instantly recognizable by region. That's why tooltip labels were put on all projection plots in the final dashboard to solve this problem.

5.3 Technical Validation

The residuals from the Bayesian model were analysed to check how well the model fitted the data. The average residual over all LADs was close to zero, showing that there was no systematic bias, while the standard deviation was 0.017. The largest residual was for the City of London, which goes along with its characterization as a statistical outlier due to the very high percentage of non-residential uses in the area. The 95% credibility intervals were tight: the actual values of 2021 were inside the predicted 2021 interval for 91.3% of LADs, which is close to the nominal 95% coverage, thus indicating that the model is only slightly overconfident.

6. Key Findings

The analysis revealed several substantive findings about socio-economic patterns across England and Wales:

- A moderate positive correlation ($r = 0.41$) was found between Level 4+ qualification rates and private renting proportions in 2021 across all 323 LADs, confirming the central hypothesis that highly qualified areas are more likely to exhibit high private renting.
- A strong negative correlation ($r = -0.39$) between Level 4+ qualifications and unemployment rate was identified, indicating that higher-qualification LADs tend to have lower unemployment, consistent with the human capital theory of labour markets.
- London boroughs dominate the top of the qualification distribution. The City of London (68.4% Level 4+), Wandsworth (53.6%), and Kensington and Chelsea (52.7%) show the highest educational attainment, and all three also show private renting rates well above the national mean.
- 322 out of 323 LADs experienced an increase in private renting between 2011 and 2021. The national mean rose by 3.5 percentage points from 15.8% to 19.3%. The City of London saw the single largest increase, rising by 12.4 percentage points.
- Bayesian forecasting projects the national mean private renting rate to reach approximately 22.8% by 2030. High-qualification London LADs are projected to see rates approaching 50% by 2030, with the City of London forecast at 51.8% (95% CI: 48.9%–54.7%).
- The PCA projection reveals a clear gradient from low-qualification, high-unemployment LADs in Wales and the North East at one end, to high-qualification, high-renting London boroughs at the other. Educational attainment appears to be a fundamental organising dimension of LAD-level socio-economic differentiation.
- The t-SNE projection reveals finer cluster structure: a tight cluster of London boroughs, a diffuse cloud of English provincial LADs, and a loose cluster of Welsh LADs with distinctively high no-qualifications rates and low private renting. The Isles of Scilly appeared as an outlier in both projections due to its extremely small population and atypical economic structure.

7. Conclusion

7.1 Socio-Economic Findings

According to this study, educational attainment strongly predicts labor market activity as well as housing tenure at the Local Authority District (LAD) level. In fact, the observation that 322 out of the 323 LADs saw an increase in private renting between 2011 and 2021 emphasizes how pervasive the changes in tenure are throughout England and Wales; on the other hand, the fact that the biggest increases have been found mostly in the highly-qualified urban areas highlights the part played by graduate worker mobility and housing affordability constraints in causing the changes [2].

The policy implications of the Bayesian projections to 2030 are quite significant. It is estimated that private renting in some London boroughs may reach or even go beyond 50% if the current situation is left unchanged by then. This would have major repercussions on the level of housing security, the ability of households to accumulate wealth, and the overall welfare of the residents in areas with a high concentration of highly qualified individuals, many of whom are renting privately not out of their lifestyle choices but out of necessity [1]. Policy measures need to be geared both towards the supply side (creating more affordable housing opportunities in areas with a high demand for graduate workers) and the demand side (removing the financial obstacles that prevent younger highly-qualified individuals from buying their own homes).

The very strong negative association between qualification level and no-qualifications rate ($r = -0.90$) is indicative of the highly polarised educational outcomes across England and Wales which have been well-documented, and it is the case that areas such as Blaenau Gwent, Knowsley, and Hull are the places where low attainment and high unemployment are both concentrated [9]. Looking at the choropleth maps

visually, it was also verified that there is a clear North-South and urban-rural difference in educational attainment which is almost the same as the pattern of private renting change.

7.2 Reflections on Information Visualisation

The project illustrated through a real-life example how each visualisation idioms can fulfill different analytic purposes. A choropleth map appears to be a powerful tool for conveying spatial patterns to laymen however it is not enough for showing the multivariate data structure. Scatter plots offer a possibility for correlation research, but in the case of 323 unlabelled points, they turn out to be very confusing. PCA and t-SNE as dimensionality reduction techniques nicely complemented the gap between the high-dimensional data environment and human-readable 2D illustration nevertheless one had to do careful labelling and provide sufficient context so that it was interpretable.

The design experience of the visualization gave Munzner's an idea that the design process is iterative by nature: the first design solutions that seemed fine (like displaying raw counts rather than proportions) were changed when they were discovered to hamper the comparison of differently-sized LADs. Also, the role of the tooltip as a way of clarifying ambiguity in scatter and projection plots became clear: without identifying individual marks, the projections would have been useful but not actionable.

Making use of Bayesian techniques enriched the work by explicitly laying out the uncertainty in the forecast, which is a more honest approach to point forecasts and it is also more handy for decision-making. There could be several ways the work extended to include different census variables (e.g. ethnicity household composition, or income deprivation indices) to create a fuller model and examine if the connection between qualifications and private renting is still there after confounding demographic factors.

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